

# Green Lab - LAB 3

R with Positron

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# 0. Setup

**0.1:** Create a new Positron Project for the lab

**0.2:** Install and load the following packages:

- *tidyverse*
- *ggplot2*
- *bestNormalize*
- *ARTool*
- *lmerTest*
- *xtable*

**0.3:** In a directory `data/Run_Table_TPCB.csv` there is data describing results of Data Analysis Tasks Benchmarking.

# 1. Data Loading and Inspection

**1.1:** Load the dataset `Run_Table_TPCCH.csv` into R.

**1.2:** Keep only the four required columns: *library*, *dataframe\_size*, *trial*, and *energy\_usage*.

**1.3:** Convert *library*, *dataframe\_size*, and *trial* into factor variables.

**1.4:** Inspect the structure of the dataset with functions such as `glimpse()`, `summary()`, and `head()` to confirm that the data is properly loaded and formatted.

## 2. Visual Exploration via Plotting

**2.1:** Create a histogram of *energy\_usage* to inspect its overall distribution and assess whether it appears normal.

**2.2:** Create boxplots of *energy\_usage* grouped by *library* and *dataframe\_size* to compare distributions across experimental conditions.

**2.3:** Create scatterplots of *energy\_usage* across *trials*. Assign different colors to the combinations of *library* and *dataframe\_size* (i.e., the “experiments”), and include a legend to distinguish them.

# 3. Data Normality and Normalization

- 3.1:** For each combination of *library* and *dataframe\_size*, run a normality test (Shapiro–Wilk).
- 3.2:** Normalize the *energy\_usage* column using the **bestNormalize** package and create a new normalized column.
- 3.3:** Plot the normalized data and repeat normality tests.
- 3.4:** Evaluate whether parametric tests can be applied after normalization.

## 4. (Non-)Parametric Tests

**4.1:** Use a non-parametric test *wilcox.test* (Wilcoxon's test) to check whether energy usage is different between libraries for small dataframes.

**4.2:** Fit an *art()* model to observe influence of *library*, *dataframe\_size* and its interactions with *energy\_used*. Treat trial as a random effect for repeated measures experiment. What can you see? Validate your model with *summary(model)*.

**4.3:** Similarly to *art*, run *lmer()* model to observe similar effects, except your target variable should be *norm\_energy\_used*.

**4.4:** Check residuals for normality and assess the validity of the model.

## 5. Summary Plots

- 5.1:** Create a boxplot to visualize the distribution of *energy\_usage* for each “experiment”. Use *facet\_wrap()* to adequately separate the experiments by library or dataframe size.
- 5.2:** Label the x-axis as “Dataframe Size” and the y-axis as “Energy Usage”
- 5.3:** Create a scatter plot (*geom\_point()*) to show how *energy\_usage* varies over trial. Assign a unique color to each “experiment” and add a legend.
- 5.4:** Using faceting, create a grid with 4 scatter plots (one per “experiment”) showing how *energy\_usage* correlates to *execution\_time*, Including trend lines.

## 6. Annotated Tables

**6.1:** Use **dplyr** functions to calculate the **average** *cpu\_usage*, *memory\_usage* and *energy\_usage* for each “experiment”.

**6.2:** Using the **xtable** package, format the summary table as LaTeX. You can copy the output for rendering in engines such as Overleaf.